

5.5

Fixed-cutoff moving-cutoff

1. statement

printed page 1104 equation (5.5) $t < t_0$

$$\rho_{X_0,t_0}(F,t) = \frac{1}{4\pi(t_0-t)} \exp\left(-\frac{|F-X_0|^2}{4(t_0-t)}\right),$$

fixed original-coordinate cutoff $\phi \in C_c^\infty(B_{2r}(X_0))$ $\phi = 1$ on $B_r(X_0)$

$$\Psi(X_0,t_0;t) = \int_{\Sigma_t} (\cos \alpha)^{-p} \phi(F) \rho_{X_0,t_0}(F,t) d\mu_t.$$

Theorem 3.3 relevant estimate

$$\frac{d}{dt} \left(e^{c_1\sqrt{t_0-t}} \Psi(X_0,t_0;t) \right) \leq -\text{nonnegative square terms} + c_2 e^{c_1\sqrt{t_0-t}}$$

right side one-sided positive error Section 5 first singular point (X_0,T)

$$F_\lambda(x,s) = \lambda(F(x,T+\lambda^{-2}s) - X_0), \quad d\mu_s^\lambda = \lambda^2 d\mu_{T+\lambda^{-2}s},$$

$w_\lambda = (\cos \alpha_\lambda)^{-p}$ printed equation (5.5) literal endpoint quantity

$$G_\lambda(s) = e^{c_1\sqrt{T-(T+\lambda^{-2}s)}} \int_{\Sigma_s^\lambda} w_\lambda \phi_R(F_\lambda) (-s)^{-1} e^{-|F_\lambda|^2/[4(-s)]} d\mu_s^\lambda,$$

ϕ_R is a compact cutoff fixed in scaled coordinates fixed $-1 < s_1 < s_2 < 0$

$$G_\lambda(s_2) - G_\lambda(s_1) \rightarrow 0 \quad (\lambda \rightarrow \infty).$$

continuation remaining moving-cutoff lemma full proof literal compact-scaled-cutoff
 lemma full Section 5 compact first-singular-flow context full-hypothesis
 counterexample fixed-original-cutoff correction

2.

status: **proven** Fixed quantity terminal limit $Q : [a,T) \rightarrow [0,\infty)$ locally absolutely continuous

$$Q'(t) \leq c_2 e^{c_1\sqrt{T-t}}$$

for almost every $t < T$

$$A(t)=\int_t^T c_2 e^{c_1 \sqrt{T-u}}\, du, \qquad M(t)=Q(t)+A(t).$$

$$A(t)\rightarrow 0 \text{ as } t\uparrow T$$

$$M'(t)=Q'(t)-c_2e^{c_1\sqrt{T-t}}\leq 0.$$

$$\begin{array}{llll} M \text{ monotone decreasing} & \text{bounded below by } 0 & M(t) & \text{finite one-sided limit} \\ M(t)-A(t) & \text{finite one-sided limit} & t_\lambda(s)=T+\lambda^{-2}s & \text{fixed } s_1,s_2<0 \end{array}$$

$$Q(t_\lambda(s_2))-Q(t_\lambda(s_1))\rightarrow 0.$$

$$\text{one-sided inequality} \quad \text{positive error term}$$

$$\begin{array}{lll} \text{status: } \mathbf{proven} & \text{fixed-cutoff rescaled endpoint theorem} & \text{original-coordinate cutoff } \chi \in \\ C_c^\infty(B_{2r}(X_0)) & \chi=1 \text{ on } B_r(X_0) & \end{array}$$

$$Q_\chi(t)=e^{c_1\sqrt{T-t}}\Psi_\chi(X_0,T;t).$$

$$\text{scaled variables} \qquad \text{expanding cutoff}$$

$$\chi^\lambda(Y)=\chi(X_0+\lambda^{-1}Y).$$

$$G_\lambda^\chi(s)=e^{c_1\lambda^{-1}\sqrt{-s}}\int_{\Sigma_s^\lambda}w_\lambda\chi(X_0+\lambda^{-1}F_\lambda)(-s)^{-1}e^{-|F_\lambda|^2/[4(-s)]}\,d\mu_s^\lambda$$

$$\text{exact identity}$$

$$G_\lambda^\chi(s)=4\pi Q_\chi(T+\lambda^{-2}s).$$

$$\text{proof sketch}$$

$$T-t=\lambda^{-2}(-s), \qquad F-X_0=\lambda^{-1}F_\lambda, \qquad d\mu_s^\lambda=\lambda^2d\mu_t.$$

$$\begin{array}{llllll} \text{K\"ahler angle} & \text{positive homothety} & \text{invariant} & \text{numerator } \omega(u,v) & \text{area denominator} & \lambda^2 \\ w_\lambda=w & \text{unnormalized scaled Gaussian factor} & & 4\pi\rho_{X_0,T} & 4\pi & \text{common normalization} \\ \text{factor} & \text{normalized kernel } [4\pi(-s)]^{-1} & & & Q_\chi(t) \text{ has a common terminal limit} & \text{epsilon} \\ \text{proof} & & & & & \end{array}$$

$$G_\lambda^\chi(s_2)-G_\lambda^\chi(s_1)\rightarrow 0.$$

$$\text{status: } \mathbf{proven} \quad \text{printed cutoff} \quad \text{fixed cutoff} \qquad \text{literal expression}$$

$$\phi_R(F_\lambda)=\phi_R(\lambda(F-X_0))$$

original coordinates moving cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

supported in $B_{2R/\lambda}(X_0)$ equal to 1 on $B_{R/\lambda}(X_0)$ λ fixed terminal-limit argument
fixed Q_χ moving family $Q_{R,\lambda}$ Q_λ nonnegative has terminal limit
 $Q'_\lambda \leq 0$

$$Q_\lambda(T + \lambda^{-2}s_2) - Q_\lambda(T + \lambda^{-2}s_1) \not\rightarrow 0.$$

individual terminal limits for a moving family are insufficient uniform bridge

status: **proven** standalone literal moving-cutoff lemma flat symplectic plane $P = \mathbb{R}^2 \times \{0\} \subset \mathbb{C}^2$
stationary flow $F(x, t) = x$ $\cos \alpha = 1$ $H = 0$ curvature/flow-energy terms vanish
 $\Sigma_s^\lambda = P$ nontrivial radial compact cutoff ϕ_R

$$I(s) = \int_P \phi_R(Y)(-s)^{-1} e^{-|Y|^2/[4(-s)]} dA(Y).$$

full uncut integral 4π compact-cut integral constant $a = -s$ change variables $Y = \sqrt{a} Z$

$$I(-a) = \int_P \phi_R(\sqrt{a} Z) e^{-|Z|^2/4} dA(Z).$$

$s_1 < s_2 < 0$ $a_1 > a_2$ radial nonincreasing cutoff transition annulus scales

$$I(s_2) > I(s_1).$$

$$G_\lambda(s) = e^{c_1 \lambda^{-1} \sqrt{-s}} I(s)$$

positive endpoint defect plane full Section 5 counterexample noncompact finite
first singular time X_0 singular point

status: **proven / conditional** actual endpoint weighted Radon measures s_1, s_2 weakly converge
nonzero two-dimensional cone measure literal compact-scaled-cutoff endpoint difference
positive limit compact continuous test function

$$Y \mapsto \phi_R(Y)(-s_i)^{-1} e^{-|Y|^2/[4(-s_i)]},$$

weak convergence integrals two-homogeneous cone radial scaling compact cutoff s -dependence
conditional obstruction full counterexample same-limit conical convergence
circularly later tangent-cone theorem

3.

wall moving cutoff order-one annular flux

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$$

fixed-cutoff calculation $D\chi_{R,\lambda}$ scales like λ $D^2\chi_{R,\lambda}$ scales like λ^2 parabolic time window $t = T + \lambda^{-2}s$ $\rho_t = \lambda^2\rho_s$ $d\mu_t = \lambda^{-2}d\mu_s^\lambda$ $dt = \lambda^{-2}ds$ powers cancel

$$\int_{s_1}^{s_2} \int_{\Sigma_s^\lambda} w_\lambda \rho_s \left[D\phi_R(F_\lambda) \cdot v_\lambda + \Delta_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) + 2\langle \nabla \log \rho_s, \nabla(\phi_R \circ F_\lambda) \rangle \right] d\mu_s^\lambda ds.$$

integral supported in fixed scaled annulus $R < |F_\lambda| < 2R$ small λ factor sign local area
 bounds fixed-time compactness zero velocity/zero curvature heuristics stationary plane
 pure spatial heat-kernel flux already equals the nonzero endpoint defect later time-independent tangent-cone
 convergence pre-(5.5) proof (5.5) estimates

4. Approach timeline

stage	question addressed	conclusion established	effect on the approach
fixed terminal quantity	Does one-sided estimate imply a finite terminal limit?	yes, by adding the integrable future error tail	validates the common-limit argument for one fixed cutoff
exact rescaling	What does the fixed original cutoff become after blow-up?	$\chi(F)$ becomes $\chi(X_0 + \lambda^{-1}F_\lambda)$, with a harmless 4π factor	gives a rigorous corrected endpoint formula
literal scaled cutoff	Is $\phi_R(F_\lambda)$ the same fixed-cutoff object?	no; it is $\chi_{R,\lambda}(F)$ in original coordinates	exposes the missing uniform moving-family step
model obstruction	Is fixed compact scaled cutoff harmless?	no; flat stationary plane gives positive endpoint defect	disproves standalone lemma, but not full compact singular-flow case
annular flux audit	Can scaling make moving-cutoff errors small?	no; leading cutoff terms are order-one on a fixed scaled annulus	identifies the exact missing bridge
fixed-time compactness	Can earlier estimates give some compactness?	yes, fixed-time local Radon compactness	insufficient to compare two endpoint measures

5. Current status & next step

literal compact-scaled-cutoff version of equation (5.5) is not proved from the available pre-(5.5) material. The strongest unconditional corrected theorem is the fixed-original-cutoff version above. No full compact first-singular-flow counterexample has been established. The remaining actionable lemma is the following uniform moving-cutoff bridge.

Remaining lemma uniform moving-cutoff bridge

Let $(\Sigma_t, F(\cdot, t))$ be a compact beta-symplectic critical surface flow smooth for $t < T$, let (X_0, T) be a singular spacetime point, fix $R > 0$, fix $\phi_R \in C_c^\infty(B_{2R}(0))$ with $\phi_R = 1$ on $B_R(0)$, and set

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

For

$$Q_{R,\lambda}(t) = e^{c_1\sqrt{T-t}} \int_{\Sigma_t} (\cos \alpha)^{-p} \chi_{R,\lambda}(F) \rho_{X_0,T}(F, t) d\mu_t,$$

let

$$L_{R,\lambda} = \lim_{t \uparrow T} Q_{R,\lambda}(t),$$

assuming this limit exists from the one-sided estimate for each fixed λ . Prove, noncircularly and uniformly at the parabolic scale, that for every fixed $-1 < s_1 < s_2 < 0$,

$$Q_{R,\lambda}(T + \lambda^{-2}s_i) - L_{R,\lambda} \rightarrow 0 \quad (i = 1, 2).$$

Equivalently, prove the annular cutoff-flux terms supported on

$$R < |F_\lambda| < 2R$$

are $o(1)$, or cancel, as $\lambda \rightarrow \infty$.